

Census of 321 Tires — Real vs Nominal Width

Constant-deflection model · BikeLab PSI

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BikeLab Studio · Trujillo, Peru · 2026 · Dataset descriptor · License CC BY-NC-ND 4.0
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Abstract

Compilation of BicycleRollingResistance's public census of 321 road and gravel tires (real vs nominal width), corrected for internal rim width. On that data the constant-deflection model (Berto 15% criterion) of the BikeLab PSI Calculator is built, cross-validated against SILCA and Rene Herse (9/12 wheels within $\pm 10\%$). Uncertainty via Latin Hypercube (10,000 samples) and Sobol indices. BikeLab's original contribution is the model, the calibration and the uncertainty analysis; the widths come from BicycleRollingResistance's public census.

Verifiable findings

Road: tires measure $\sim +0.8$ mm over nominal on an 18 mm internal rim. Gravel: ~ -1.8 mm under nominal. Sobol: on road real width dominates (S1 up to 0.68); on gravel caliper gauge error dominates (S1 up to 0.73). Rider weight explains under 9%. ETRTO hookless caps: 72.5 PSI (60 PSI at 30 mm internal or more).

Sample

Case	Configuration	Mean [p5-p95]	Band +/-PSI
1	83 kg · 28 mm · road · rim 21	73 [63-86]	11.7
2	95 kg · 28 mm · road · rim 21	84 [72-100]	13.8
3	75 kg · 32 mm · road · rim 19	63 [54-73]	9.2
4	85 kg · 40 mm · gravel · rim 24	38 [35-41]	3.2
5	100 kg · 45 mm · gravel · rim 25	39 [36-41]	2.5
6	70 kg · 25 mm · road · rim 21	71 [59-85]	12.8

Dataset variables

nominal width (mm); measured real width (mm); internal rim width (mm); recommended pressure (PSI); uncertainty band ($\pm$ PSI). The width census is public (BicycleRollingResistance). What is shared under agreement is what BikeLab adds: the correction tables by model/rim, the calibrated coefficients and the sampling code.

Data source and acknowledgment

The real-vs-nominal width census (159 road and 162 gravel tires) comes from the public database of BicycleRollingResistance (Jarno Bierman), bicyclerollingresistance.com, used with attribution. We thank BRR for publishing open, rigorous data that makes this model possible.

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Provisional by design. The width census is public data from BicycleRollingResistance, used with attribution; the correction tables, coefficients and code are shared under agreement. Contact: carlos.ravello@zoovettravel.com